Roll No.: ....

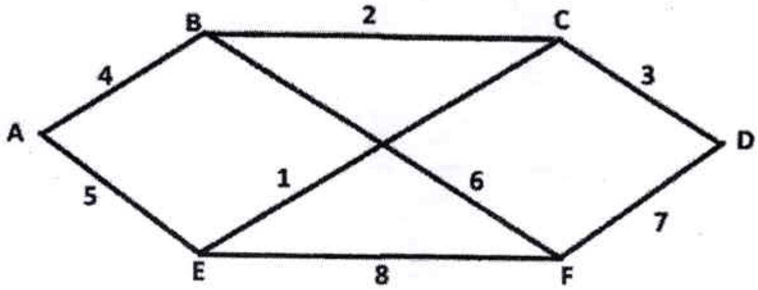
END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech	Semester: VI
Name of the Course: Computer Networks-I	Course Code: TCS-604
Time: 3 Hours	**Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	List two ways in which the OSI reference model and the TCP/IP reference model are the same. Now list two ways in which they differ.	CO1
(b)	What access network technologies would be most suitable for providing internet access in rural areas? Discuss in detail.	CO1
(c)	A user can directly connect to a server through either long-range wireless or a twisted-pair cable for transmitting a 1500-bytes file. The transmission rates of the wireless and wired media are 2 and 100 Mbps, respectively. Assume that the propagation speed in air is 3×10^8 m/s, while the speed in the twisted pair is 2×10^8 m/s. If the user is located 1 km away from the server, what is the nodal delay when using each of the two technologies?	CO1
Q2		
(a)	Electronic mail systems need directories so people's e-mail addresses can be looked up. To build such directories, names should be broken up into standard components (e.g., first name, last name) to make searching possible. Discuss some problems that must be solved for a worldwide standard to be acceptable.	CO3
(b)	Why is a content provider considering a different Internet entity today? How does a content provider connect to other ISPs? Why?	CO2
(c)	What are the various DNS record types? How it works? Discuss Root, Top-Level and Authoritative DNS servers.	CO3
Q3		
(a)	Suppose that the TCP congestion window is set to 16 kb and a time out occurs. How big will the window be if the next four transmission bursts are all successful? Assume that the maximum segment size (MSS) is 1 kb.	CO4
(b)	Show the FSM description for rdt3.0 sender. How is it different from rdt2.2 sender?	CO4
(c)	Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose propagation time from the earth is 270 msec. Acknowledgements are always piggybacked onto data frames. The headers are very short. Three-bit sequence numbers are used. What is the maximum achievable channel utilization for Stop-and-wait protocol and Go Back N protocol.	CO4
Q4		
(a)	Consider the subnet of Fig. 1-a, Distance vector routing is used, and the following vectors have just come into router C: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0,	CO5

	9, 10); and from E: (7, 6, 3, 9, 0, 4). The measured delays to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the expected delay.  **Fig. 1-a**	
(b)	Write down the name of three strategies used to handle the transmission from IPv4 to IPv6?	CO5
(c)	What is the principal difference between virtual circuits and datagram networks? Give an example.	CO5
Q5		
(a)	A bit stream 10011101 is transmitted using the standard CRC method described in the text. The generator polynomial is x^3+1. Show the actual bit string transmitted. Suppose the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end.	CO6
(b)	What is the difference between a pure ALOHA and a slotted ALOHA?	CO6
(c)	Consider a network path P—Q—R between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path by splitting the file into chunks of 10^3 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU. Each of the links P—Q and Q—R has a bandwidth of 10^6 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link P—Q and transmit on link Q—R. Assume P starts transmitting the chunks at time $t = 0$. How long it takes for node R to receive all the chunks of the file sent by node P?	CO2